

AI Smart Device Control & Automation Assistant

STEP 05 PROJECT DOCUMENTATION — Bluetooth Device Discovery

Item	Validated Project Details
Completed Steps	01 — Initialization; 02 — Configuration & Command Model; 03 — Command Parser; 04 — Laptop Automation; 05 — Bluetooth Discovery
Platform	Windows / PowerShell
Python	3.11.9 — venv
Bluetooth Library	Bleak
Automated Tests	11/11 PASS — 100%
Live Discovery	60 Bluetooth devices discovered
Step Status	COMPLETED — PASS

1. Executive Summary

Step 05 implemented the Bluetooth device discovery layer using the Bleak library. The discovery module scans nearby Bluetooth Low Energy devices, converts each result into a simple structured record containing a device name and address, and displays the discovered devices. Live validation completed successfully and returned **60 devices**. The complete project automated test suite collected 11 tests and passed all 11 in **0.20 seconds**.

2. Bluetooth Discovery Architecture

```
Windows Bluetooth Adapter ↓ Bleak ↓ BleakScanner.discover() ↓ BluetoothDiscovery ↓  
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ ■ Device Name ■ ■ Device Address ■  
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ ↓ Structured Device List
```

The Step 05 module is discovery-only. Device connection, disconnection, and device control are intentionally reserved for later project steps.

3. Bluetooth Library Selection

The original PyBluez installation path was tested but was not compatible with the project environment. PyBluez 0.23 failed during installation with the setup error **use_2to3 is invalid**. PyBluez2 was then attempted and failed because the source package reported **package directory 'bluetoothwindows' does not exist**. The project therefore moved to Bleak for Windows Bluetooth Low Energy discovery.

Library Attempt	Result	Observed Issue
PyBluez 0.23	Failed	use_2to3 is invalid
PyBluez2 0.46	Failed	bluetoothwindows package directory missing
Bleak	Successful	Bluetooth discovery completed successfully

4. Discovery Module

The main module is **src/bluetooth_discovery.py**. It uses an asynchronous **BluetoothDiscovery** class. The **discover_devices()** method calls **BleakScanner.discover()**, handles discovery exceptions, and converts returned devices into dictionaries containing **name** and **address**. The **display_devices()** method formats the result for terminal inspection.

5. Live Bluetooth Discovery Result

A live discovery scan was successfully executed on the Windows system. The scan completed and displayed device records through item 60. The observed result therefore confirms **60 Bluetooth devices discovered** during the validation scan.

```
Example observed records: 55. Name : Unknown Device Address : 33:5C:E8:B2:75:3F 56. Name : Unknown Device
Address : 3F:79:D5:8C:42:73 ... 60. Name : Unknown Device Address : C1:22:33:55:6A:FD ■ Bluetooth
discovery operation completed.
```

Some discovered devices were reported as **Unknown Device** while still providing an address. This is accepted by the Step 05 discovery layer; friendly-name resolution and device selection can be improved in later steps.

6. Automated Testing — Final Result

The complete project test suite was executed from the project root after adding the Step 05 discovery tests. All previous Step 03 and Step 04 tests remained passing, and all three new Bluetooth discovery tests passed.

Test	Result
test_bluetooth_discovery_initialization	PASSED
test_device_record_structure	PASSED
test_display_empty_device_list	PASSED
test_turn_on_light	PASSED
test_turn_off_fan	PASSED
test_open_chrome	PASSED
test_bluetooth_devices	PASSED
test_unknown_command	PASSED
test_laptop_controller_initialization	PASSED
test_non_laptop_command_is_rejected	PASSED
test_unsupported_application_is_rejected	PASSED
Total	11 passed in 0.20s

7. Test Coverage Summary

Project Area	Tests	Status
Command Parser	5	5/5 PASS
Laptop Controller	3	3/3 PASS
Bluetooth Discovery	3	3/3 PASS
Overall	11	11/11 PASS — 100%

8. Privacy / Documentation Note

Bluetooth addresses can identify individual physical devices. Real addresses from the live scan should therefore be masked in public GitHub documentation. The project report records the successful discovery result without requiring real device addresses to be published.

9. Troubleshooting / Lessons Learned

Issue	Resolution / Lesson
PyBluez installation failure	Stopped using the legacy package after the use_2to3 setup error.
PyBluez2 installation failure	Stopped using the alternative after the missing bluetooth/windows package error.
Library migration	Moved Step 05 discovery to Bleak for the Windows Bluetooth LE workflow.
Unknown device names	Discovery accepts records with an unavailable friendly name and retains the address.
Hardware-dependent testing	Automated tests validate module behavior and record structure; live Bluetooth scanning is validated separately.

10. Step 05 Milestone Status

Milestone	Status
Bluetooth library installed	COMPLETED — Bleak
Discovery module	COMPLETED
Device record structure	COMPLETED
Empty-list handling	COMPLETED
Live Bluetooth scan	PASS — 60 devices
Automated test suite	11/11 PASS — 100%

11. Project Status After Step 05

Step	Milestone	Status
01	Project Initialization & Environment Setup	COMPLETED
02	Configuration & Device Command Model	COMPLETED
03	Command Parser / Intent Model	COMPLETED — 5/5 PASS
04	Laptop Automation Foundation	COMPLETED — 8/8 PASS + Notepad PASS
05	Bluetooth Device Discovery	COMPLETED — 11/11 PASS + 60-device scan
06	Bluetooth Device Control	Pending
07	IoT Control Foundation	Pending
08	AI Intent Integration	Pending
09	Voice Input Integration	Pending
10	Voice Response Integration	Pending
11	Command Router	Pending
12	Activity Logging	Pending
13	Error Handling	Pending
14	Dashboard Foundation	Pending
15	Dashboard Integration	Pending
16	Automated Testing	Pending — final suite later
17	End-to-End Testing	Pending
18	Performance Evaluation	Pending
19	Documentation & Final Report	Pending
20	Final Demonstration & Closure	Pending

12. Step 05 Conclusion

STEP 05 — BLUETOOTH DEVICE DISCOVERY: PASS. The project now has a working Bluetooth Low Energy discovery layer based on Bleak. The live scan successfully discovered 60 device records, and the complete automated test suite passed all 11 tests in 0.20 seconds. Bluetooth connection and control are intentionally left for Step 06.

13. Revision Record

Revision	Update
STEP 01	Project initialization, environment setup, validation, roadmap, and documentation foundation.
STEP 02	Configuration, configuration loader, DeviceCommand model, validation, and milestone status.
STEP 03	Deterministic command parser, intent mappings, unknown-command handling, package/import correction, and 5/5 PASS.
STEP 04	LaptopController, application allowlist, parser integration, 8/8 automated tests PASS, and successful Notepad launch.
STEP 05	Added Bleak Bluetooth discovery, handled PyBluez/PyBluez2 compatibility failures, validated live discovery of 60 devices, and achieved 11/11 automated tests PASS in 0.20s.
Next	Continue with Step 06 — Bluetooth Device Control and update this same report after validation.